Safe Autopilot

What Could Go Wrong, and How I Tested It

Godwin Pammichukwu Emefiele Jr

github.com/donpams/safe-autopilot | JSBSim + Gymnasium + PyTorch + Stable-Baselines3

A learned flight controller with a runtime safety monitor that falls back to a PID whenever the aircraft's state leaves a validated safe envelope, tested end-to-end in a Cessna 172 JSBSim simulation. v1 learns by imitating the PID; v2 learns by reinforcement learning (PPO), behind the same monitor and fallback. Below are the seven real issues found during development, each with its root cause, how it was caught, and its resolution or documented scope boundary.

0 of 180	**2–3×**	**Sev. 3**
monitored flights crashed under injected faults (BC alone crashed in 100% of all three fault tests)	better final altitude tracking with the monitor engaged (v1 stress test)	fallback's reliability cliff, shared by every controller tested, RL included

Seven things that broke, and what I learned from each

Issue	Root cause	Caught by	Resolution
Heading-wrap discontinuity	Heading crossing 360°/0° gave a 358° error instead of 2°, commanding a hard turn the wrong way.	Reading the first hands-off trim plot, before writing any controller.	Wrapped heading error to [-180°, 180°) inside the environment, before any controller sees it.
Deterministic turbulence	JSBSim's gust model defaults to a fixed seed, so every training episode got the identical disturbance.	Tracing raw state across episode seeds and finding identical values.	Randomized atmosphere/randomseed on every episode reset.
Imitation model loses integral memory	The PID's altitude loop integrates error; the cloned model sees only instantaneous state, so small errors compound over the ~30 s phugoid mode into a slow oscillation.	Extending the evaluation horizon to 120 s after the model looked fine at 60 s.	Not fixed: it is the core reason for the safety monitor, and for moving to RL in v2.
PID fallback mis-trims when engaged mid-flight	The PID took the pitch it saw on its first call as “level”: right at reset, wrong when the monitor engaged it mid-oscillation.	Integration-testing the full monitor + fallback system; invisible to a PID-only unit test.	Environment exposes the true trim pitch; the PID takes it explicitly.
Fallback has no airspeed protection	The PID controls only elevator and aileron. A bank plus a gust can bleed airspeed through the 50 kt floor with nothing to stop it.	Systematic stress sweep over turbulence severity, independent of model and monitor.	Documented limit: reliable to severity 2, 50–65% crashes at severity 3–4, for every controller. Needs a throttle loop.
RL reward taught the policy to crash on purpose	Unbounded per-step penalty plus a flat −50 crash penalty: once a rollout drifted, ending the episode was cheaper than continuing. Mean episode length collapsed from ~400 to ~25 steps.	Reading the full training curve, not just that the run finished. My reward check covered a perfect flight and an instant crash, but not a slow divergence.	Capped the per-step penalty and added a per-step alive bonus. Episodes returned to full length (1200 steps) and reward rose.
Results depend on the simulator version	On JSBSim 1.3.3 the PID alone never crashed at severity 3–4; on 1.2.4 it crashes 50–65% of the time.	Re-running the stress test in a fresh environment with a newer JSBSim: it failed to reproduce my own earlier numbers.	Pinned jsbsim==1.2.4; every reported result uses it. RL policies were trained on 1.3.3 (stated in the README).

Validation detail

v1 stress test (20 flights/condition, 90 s, turbulence ≤ severity 2, offsets to ±150 ft, sensor noise): mean final altitude error, monitor vs. model alone – large off-nominal start 18.7 ft → 6.2 ft; combined stress 17.1 ft → 7.5 ft. No monitored flight crashed. Run against the PID itself in 7 nominal cases, the monitor stayed quiet in 4; the other 3 started beyond its 35 ft altitude limit and tripped at t=0 by design.

v2 (PPO, 2 seeds, JSBSim 1.2.4, 20 identical flights/cell): final altitude error beats the imitation model in every condition and the PID in every disturbed one (gusts: 4.7–5.7 ft vs. PID 8.1, BC 18.6); the PID is still best in calm air (1.4 ft vs. 2.8–3.8). **Fault injection** (elevator bias +0.4, stuck aileron; learned controller only): 0 of 180 monitored flights crashed, caught in about 1–1.5 s. Unmonitored, the imitation model crashed in 100% of both fault types and the RL policies in 100% of stuck-aileron flights.

Fallback reliability limit (alone, turns + turbulence): PID 0% crash at severity 0–2, 50% at 3, 65% at 4; RL policies 55–75% at severity 3–4. **Open issue:** the RL policy banks past the monitor's 30° limit and trips it in 50–60% of calm-air flights, so thresholds tuned for v1 are conservative for RL.

Scope: simulation only; two training seeds and 20 flights per cell, so small differences in calm conditions are within noise. Fault types and sizes were chosen by me. No throttle control, so the turbulence cliff above severity 2 remains open.